

Wei-Jhih Huang

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Education

Bachelor, Communication Engineering, National Central University

2022 – 2026

GPA: 3.94/4.3

Taoyuan, Taiwan

- Research Interests: AI-Native 6G Networks, Intelligent Wireless Resource Orchestration, Edge Intelligence and Computing

Research Experience

Wireless Mobile Network Lab, National Taiwan University

Apr 2026 – Present

Research Assistant, Advisor: **Prof. Hung-Yu Wei**

Taipei, Taiwan

- Explored an AI-native orchestration framework for LLM deployment across distributed edge networks, resolving end-to-end latency by strategically balancing communication overhead with computational resource availability.
- Designed predictive resource management algorithms within a 6G AI-RAN architecture, utilizing cross-layer telemetry to anticipate network dynamics and enable proactive data staging for seamless service continuity in highly mobile scenarios.

Information Processing and Communications Lab, National Central University

Feb 2024 – Jun 2026

Undergraduate Research Student, Advisor: **Prof. Chih-Wei Huang**

Taoyuan, Taiwan

- Investigated a self-aware Meta-Reinforcement Learning architecture for adaptive 6G network management, designing robust Out-of-Distribution detection mechanisms to control online fine-tuning stability under uncertain traffic shifts.
- Developed a generative model based trajectory-embedding mechanism for scenario distribution-aware DRL fine-tuning, reducing unnecessary DRL retraining by **90%** and improving user quality of service by **11%** in cross-layer resource management scenarios.

Electronic Design Automation Lab, The Chinese University of Hong Kong

Jul – Aug 2025

Summer Visiting Research Student, Advisor: **Prof. Bei Yu**

Hong Kong

- Enhanced *NUA-Timer++* for graph-based hierarchical circuit timing prediction under Non-Uniform Input Arrival Times, achieving a **1.96x** improvement in consistency and reducing average error to **0.42x** compared to SOTA.
- Collaborated with a postdoctoral researcher on Transformer architecture design and loss function refinement for *NUA-Timer++*; independently investigated a critical-path-based timing constraint extension as an auxiliary signal for logic synthesis.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management," *In Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- Ziyi Wang, **Wei-Jhih Huang**, Fangzhou Liu, Tsung-Yi Ho, David Z. Pan, Bei Yu. "NUA-Timer++: Timing Prediction for Hierarchical RTL under Non-Uniform Input Arrival Times," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, Under Review.
- Ching-Wei Lin, **Wei-Jhih Huang**, Sheng-Liang Wu, Chih-Wei Huang, Phone Lin, Shun-Ren Yang. "PACI-Net: Physics-Aware Distillation for Explainable Radio Resource Management," *IEEE Global Communications Conference (GLOBECOM)*, Under Review.

Honors & Awards

- **Outstanding Student Scholarship**, National Central University, 2026
- **Undergraduate Research Fellowship**, National Science and Technology Council, Taiwan, 2025–2026
- **Honorable Mention**, Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025

Work Experience

Chunghwa Telecom

Sep – Dec 2025

AI Software Development Intern

Taipei, Taiwan

- Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by **33%**.

Industrial Technology Research Institute

Jun – Sep 2024

AI Software Summer Intern

Hsinchu, Taiwan

- Built and deployed an LLM-based service chatbot at *Taipei Songshan Sports Center*, integrating Azure OpenAI with Linux backend services to provide **24/7** automated customer support, reducing customer queue time by **50%** and improving service efficiency.